OM GOVIND JHA

omgovindjha@iisc.ac.in | omgovinds15@gmail.com | +91 8789470367 | Bengaluru, India
github.com/OmGovind15 | LinkedIn | Portfolio Website

Education

Indian Institute of Science (IISc) — Bengaluru, India
M.Tech. in Robotics and Autonomous Systems — *2026 – 2028*

Indian Institute of Science Education and Research (IISER) — Bhopal, India
B.S. in Computer Science and Engineering, Minor in Data Science; ***CPI: 8.72 / 10.0*** — *2022 – 2026*

Technical Skills

Languages & Systems: Python, C/C++, SQL, Bash, Git, Linux, Docker, CUDA, LaTeX
ML Frameworks & Libs: PyTorch, Torchvision, Flower (Federated Learning), Hugging Face, OpenCV, YOLOv8, Scikit-learn, NumPy, Pandas, SHAP
Deep Learning & AI: Flow Matching, Generative Diffusion, Vision Transformers (ViT), Clustered Federated Pruning, Multi-Instance Learning, CNNs, BiLSTM
Math & Optimization: Convex Optimization, Optimal Transport, Statistical Inference (MLE, Hypothesis Testing), Game Theory

Publications & Preprints (Primary / First Author on all 4 works)

Modeling Affective Distress from Video: A Modality-Aware Multi-Branch Learning Approach | ***ACM DL*** — Oct 2026
Om Govind Jha (1st), N. K. Sahu, H. R. Lone. *Proceedings of 28th ACM International Conference on Multimodal Interaction (**ICMI '26**).*

Teacher-Guided Discriminative Flow Matching for Latent Representation Learning | *NLDL 2027* — Accepted
Om Govind Jha (1st), M. Bamniya, A. Borthakur. *Northern Lights Deep Learning Conference (**NLDL 2027**) – To Appear.*

CA-AFP: Cluster-Aware Adaptive Federated Pruning for Heterogeneous FL | *CoDS-COMAD 2026* — Under Review
Om Govind Jha (1st), H. Shukla, H. R. Lone. *ACM IKDD Conference on Data Science and Management of Data (**CoDS-COMAD 2026**).*

Predictive Modeling in ODI Cricket: Training Strategy, Feature Engineering, & Explainability — Under Review
Om Govind Jha (1st), K. Kandhway. *International Journal of Sports Science & Coaching (**SAGE Publications**) – Revision Stage.*

Research & Engineering Experience

Research Intern | *Supervisor: Dr. Ayon Borthakur* — May 2025 – Mar 2026
Indian Institute of Technology (IIT) Guwahati — *Guwahati, India*
- Engineered a teacher-guided Discriminative Flow Matching (DFM) framework using independent vector-field predictors on shared backbones.
- Formulated image classification as a continuous probability transport path; boosted CIFAR-100 top-1 accuracy to 69.38% and Tiny-ImageNet to 63.50%.
- Designed block-wise local training for Vision Transformers (ViT) and object detectors, cutting peak GPU memory relative to backpropagation.

Research Intern | *Supervisor: Dr. Kundan Kandhway* — Nov 2024 – Mar 2025
Indian Institute of Science Education and Research (IISER) Bhopal — *Bhopal, India*
- Developed an end-to-end sports analytics pipeline over 1,394 international ODIs; engineered 11 dynamic metrics (Pressure Index, Target Deficit).
- Enforced a chronological expanding-window training strategy to eliminate temporal data leakage in real-world sports forecasting.
- Trained 7 ML architectures with over-wise model switching, achieving 84.6% 2nd-innings accuracy; applied SHAP for game-theoretic feature attribution.

Summer Research Fellow | *Supervisor: Dr. Rajesh Sundaresan* — May 2024 – July 2024
Indian Institute of Science (IISc), Bengaluru — *Bengaluru, India*
- Formulated convex optimization frameworks using Fisher Information Matrices and Maximum Likelihood Estimation for budget-constrained serosurveys.
- Analyzed minimax game formulations for worst-case disease prevalence bounds and multi-group population test budget allocation.

Selected Projects

MAMBL: Modality-Aware Multi-Branch Learning for Video Distress Detection — Dec 2025 – May 2026
- Architected a dual-branch MIL pipeline routing sparse Action Units to 1D CNNs and continuous pose/gaze dynamics to BiLSTMs.
- Integrated gated temporal attention and gradient-based spatial saliency, achieving 71.04% accuracy on clinical Social Anxiety Disorder video benchmarks.

Real-Time Vehicle Detection and Speed Tracking | *YOLOv8, OpenCV, Python* — Sep 2024 – Nov 2024
- Implemented an automated traffic monitoring system using YOLOv8 with perspective transformation for metric speed estimation and overspeed violation alerting.

Honors, Fellowships & Leadership

Honors: Reliance Foundation Scholar (2022–26); INSA-IASc-NASI Summer Fellow (2024); GATE DA Qualified (2025); JEE Advanced (2022); AIR 98 IMU-CET.
Service: Teaching Assistant, *Data Science in Practice* (Jan–May 2025); Coordinator, Economics Club, IISER Bhopal (2024–25).